

Chapter 5

Sequence-to-Sequence with Attention

김지환

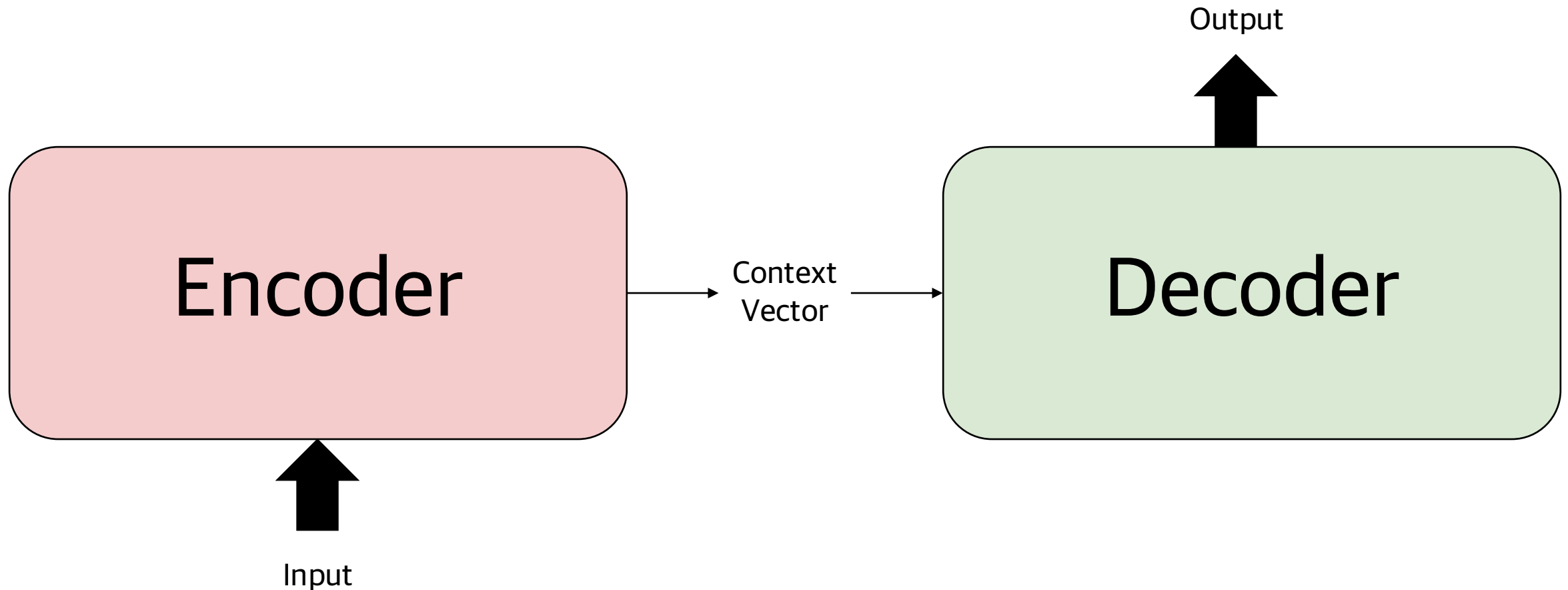
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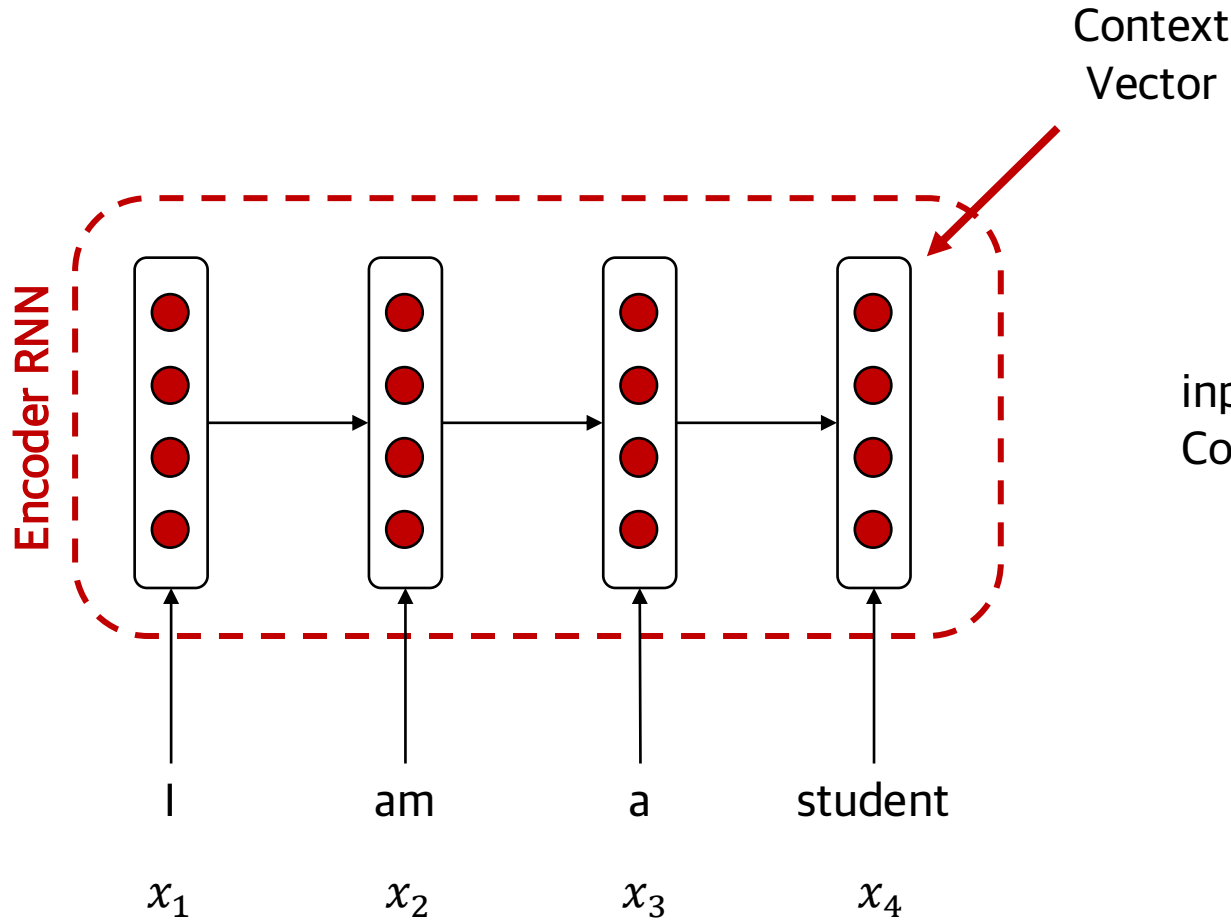
5.1 Seq2Seq

5.2 Attention

5.1 Seq2seq : Encoder-Decoder Model

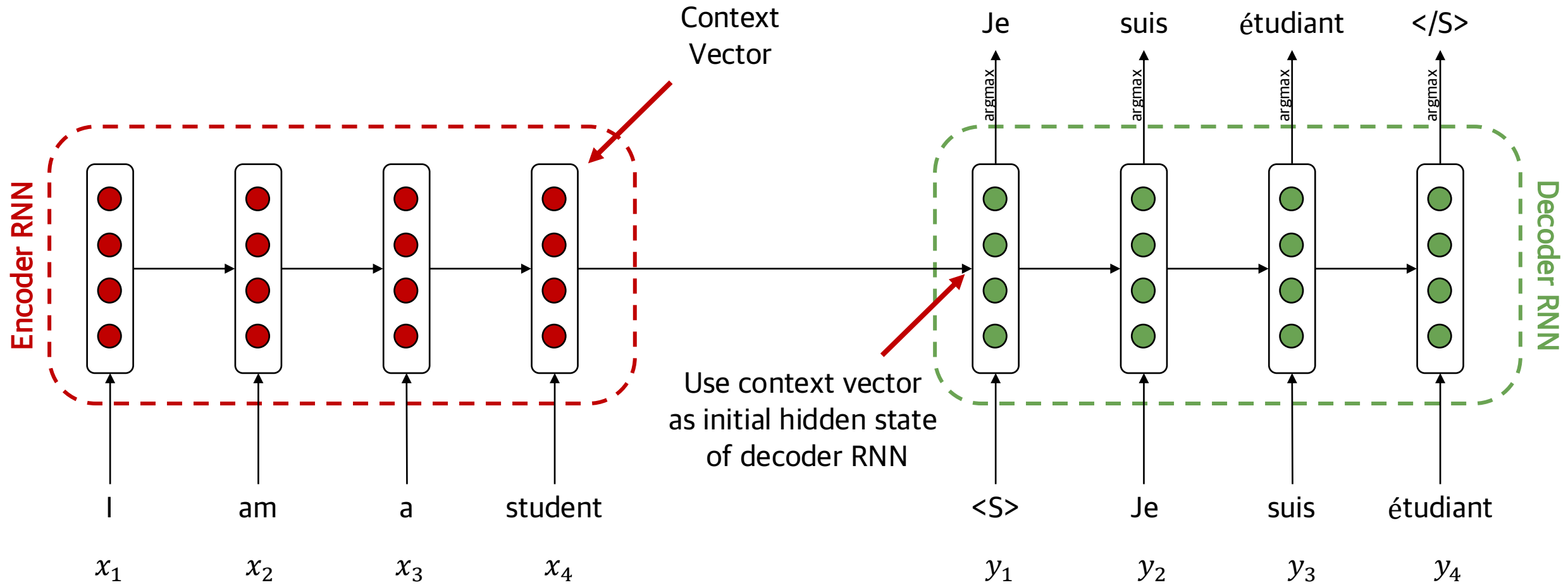


5.1.1 Seq2seq : Encoder

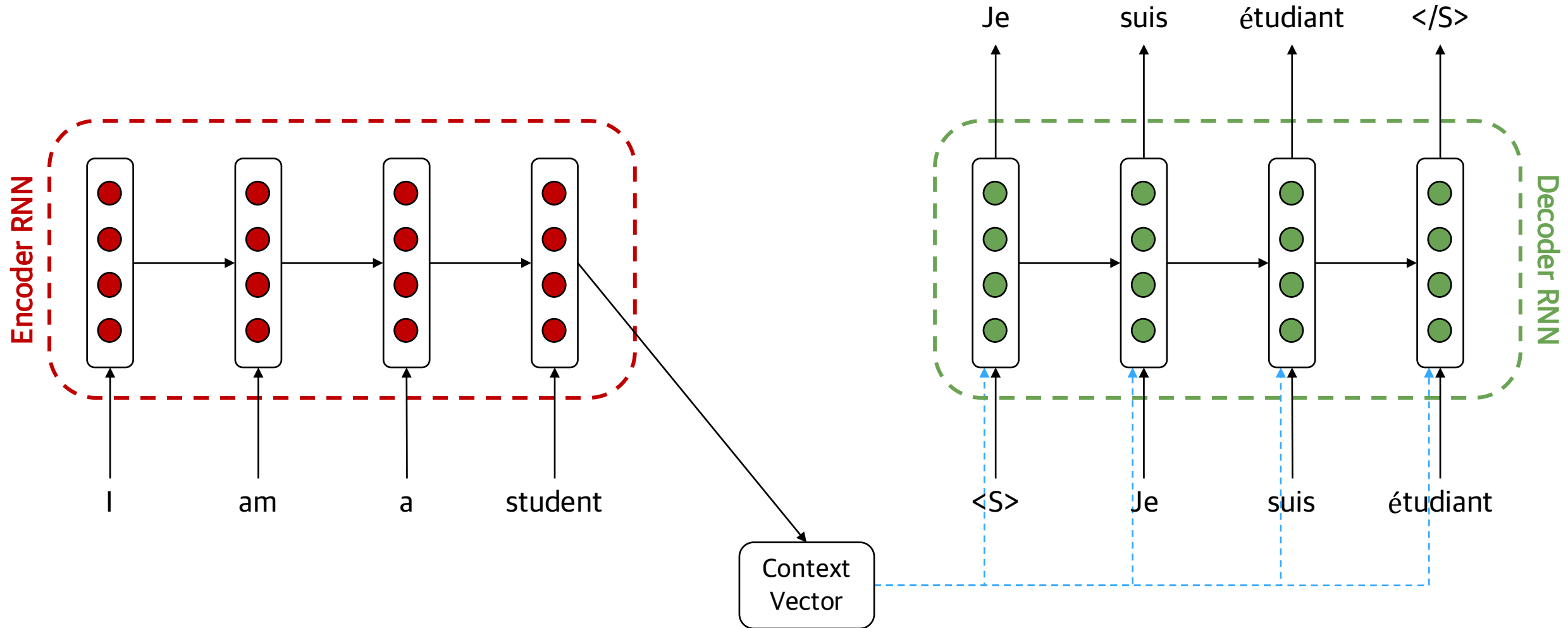


input token $x_1, \dots, x_T$ 에 대하여
Context vector C 는 T 시점에서의 RNN의 hidden state, 즉 h_T

5.1.2 Seq2seq : Decoder



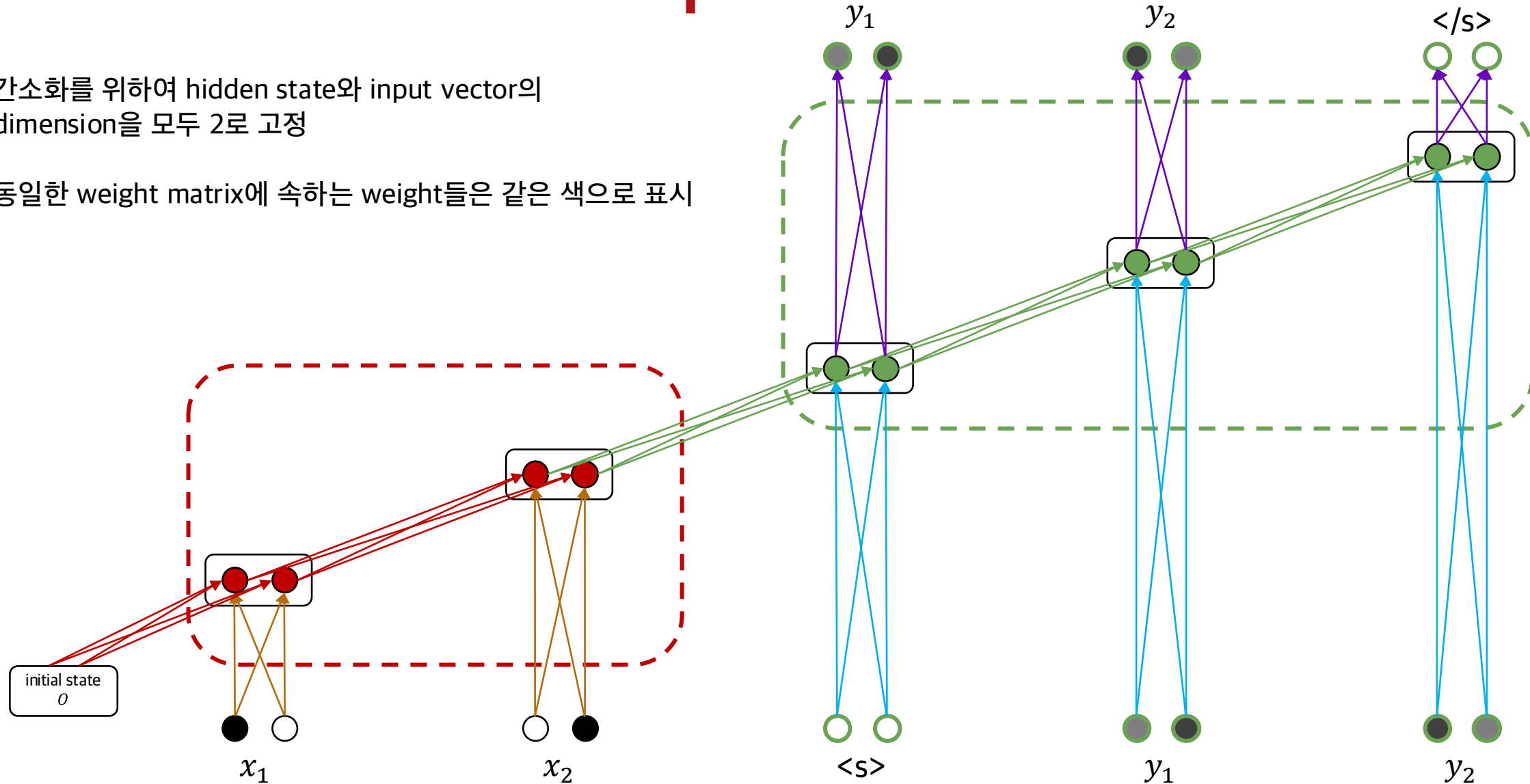
5.1.2 Seq2seq : Decoder (Cho *et al.*)



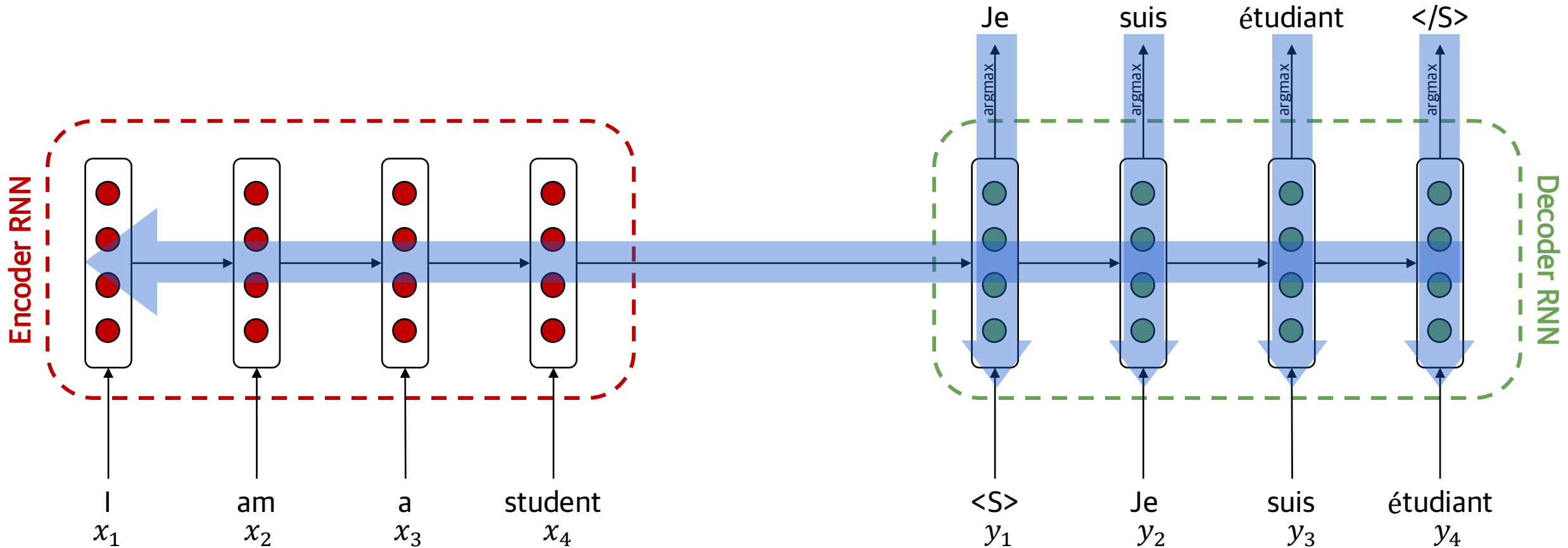
5.1.3 Seq2seq : Encoder-Decoder Graph Representation

간소화를 위하여 hidden state와 input vector의 dimension을 모두 2로 고정

동일한 weight matrix에 속하는 weight들은 같은 색으로 표시

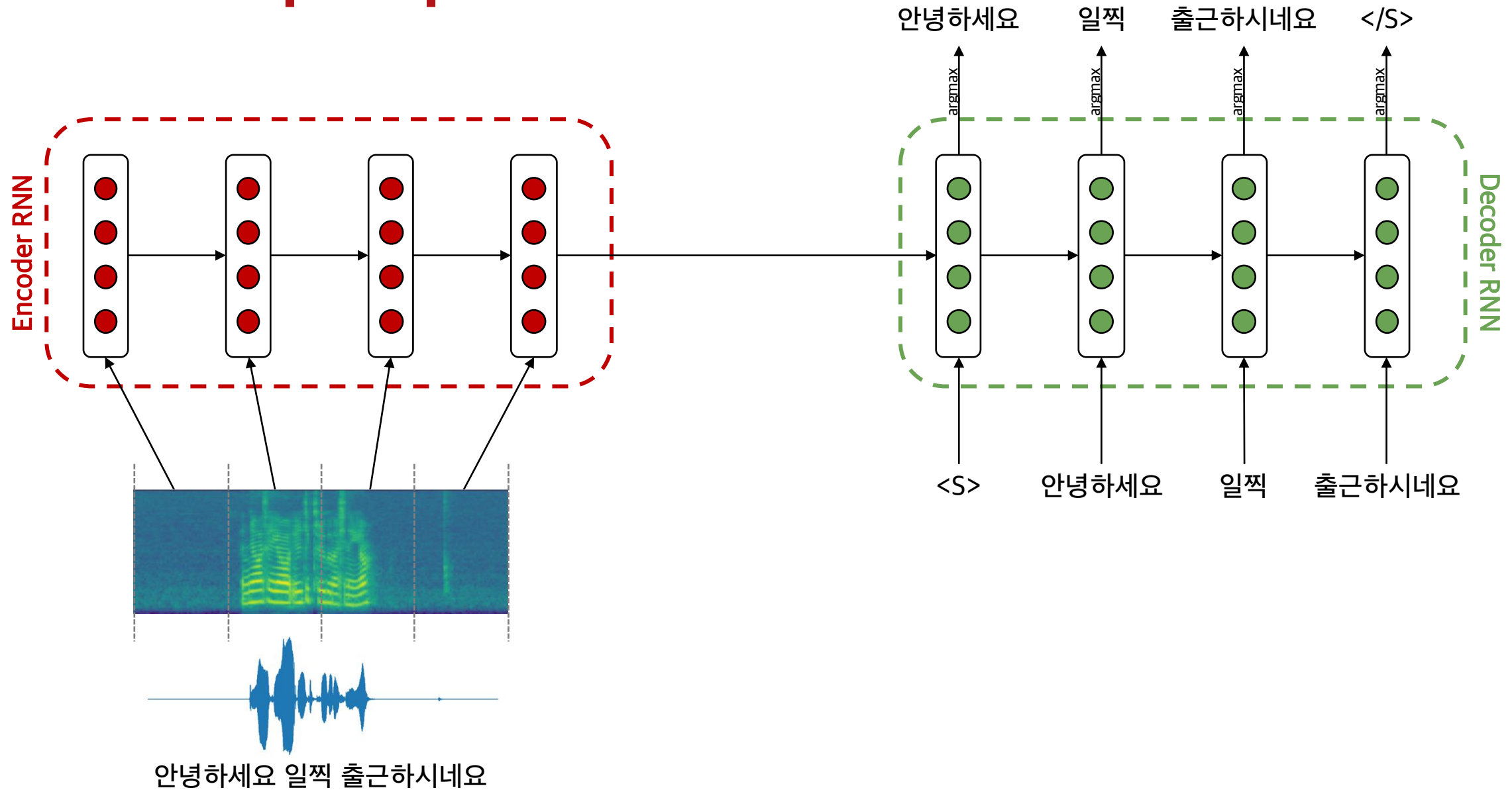


5.1.3 Seq2seq : Encoder-Decoder Model Training

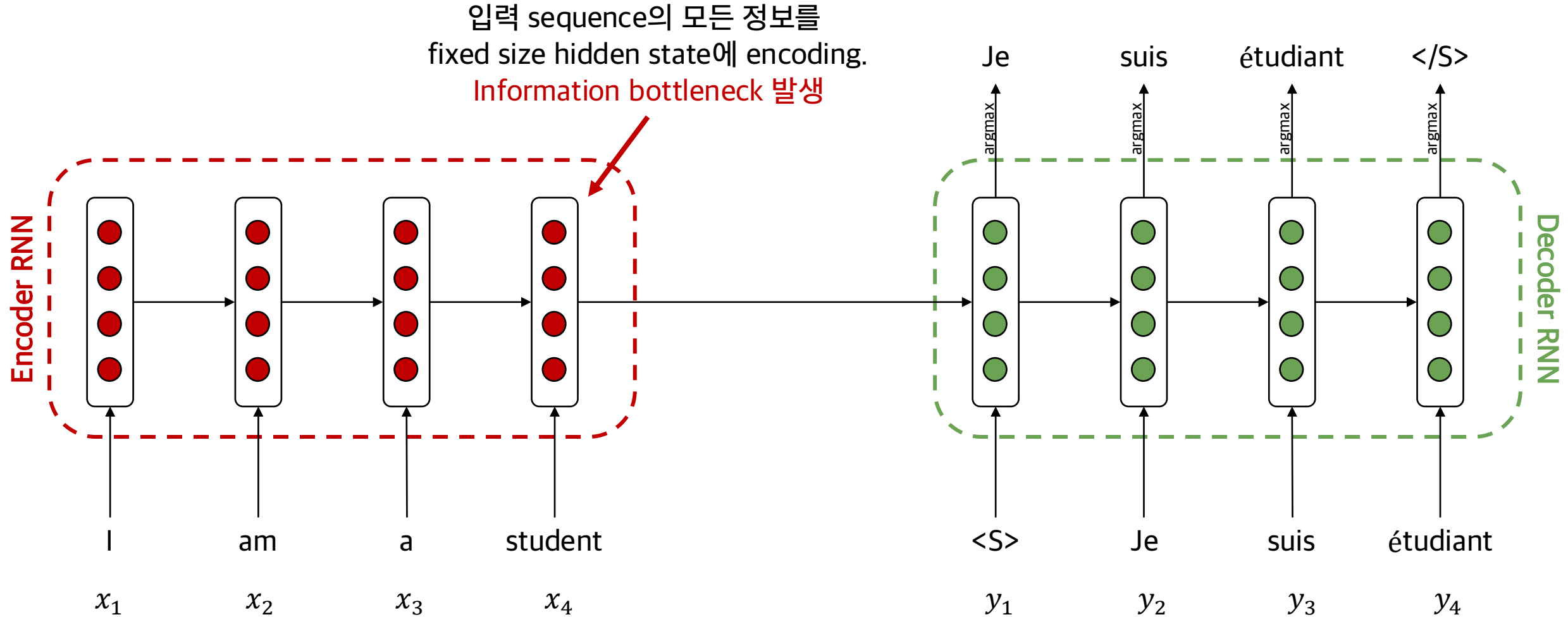


Encoder-decoder is optimized as a single system. Backpropagation operates “end-to-end”

5.1.4 Seq2seq : Encoder-Decoder Model for ASR



5.2 Seq2Seq : The Bottleneck Problem

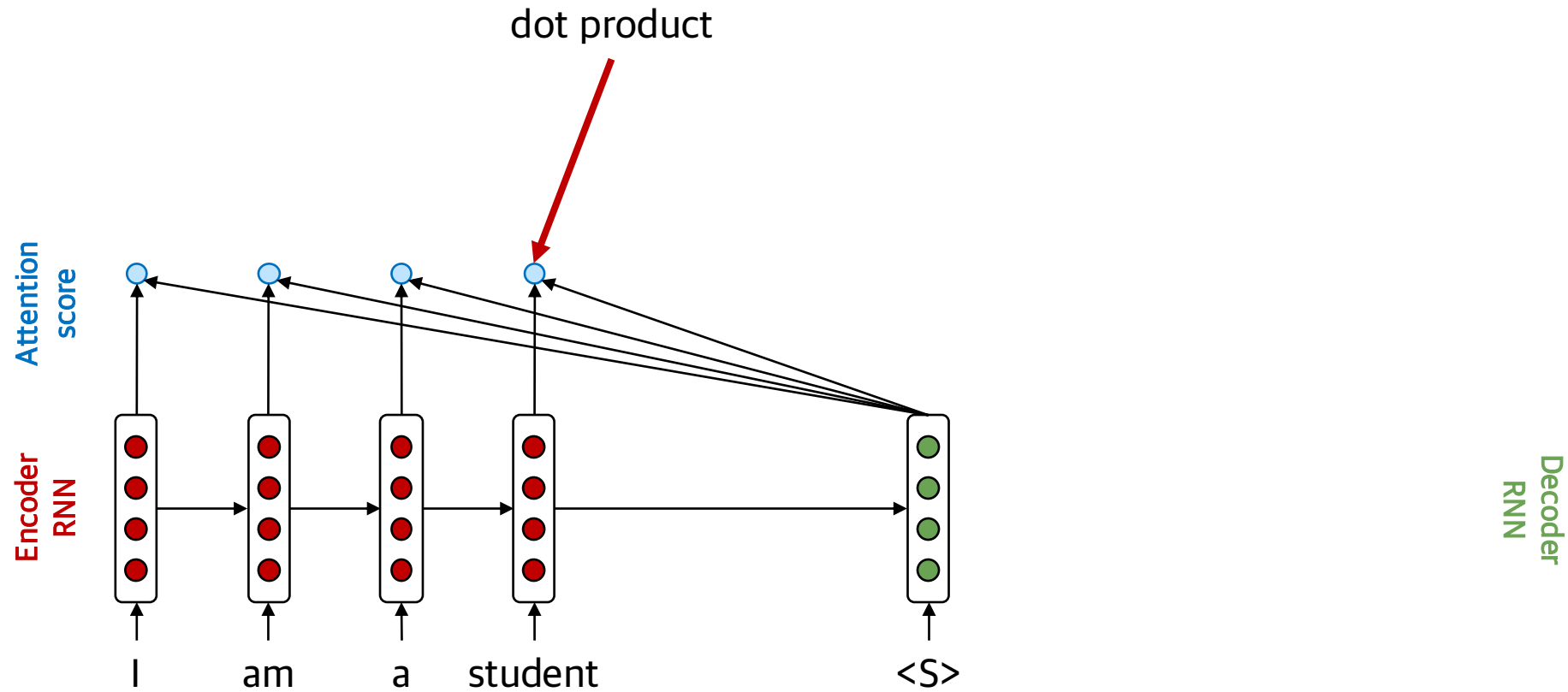


5.2.1 Attention

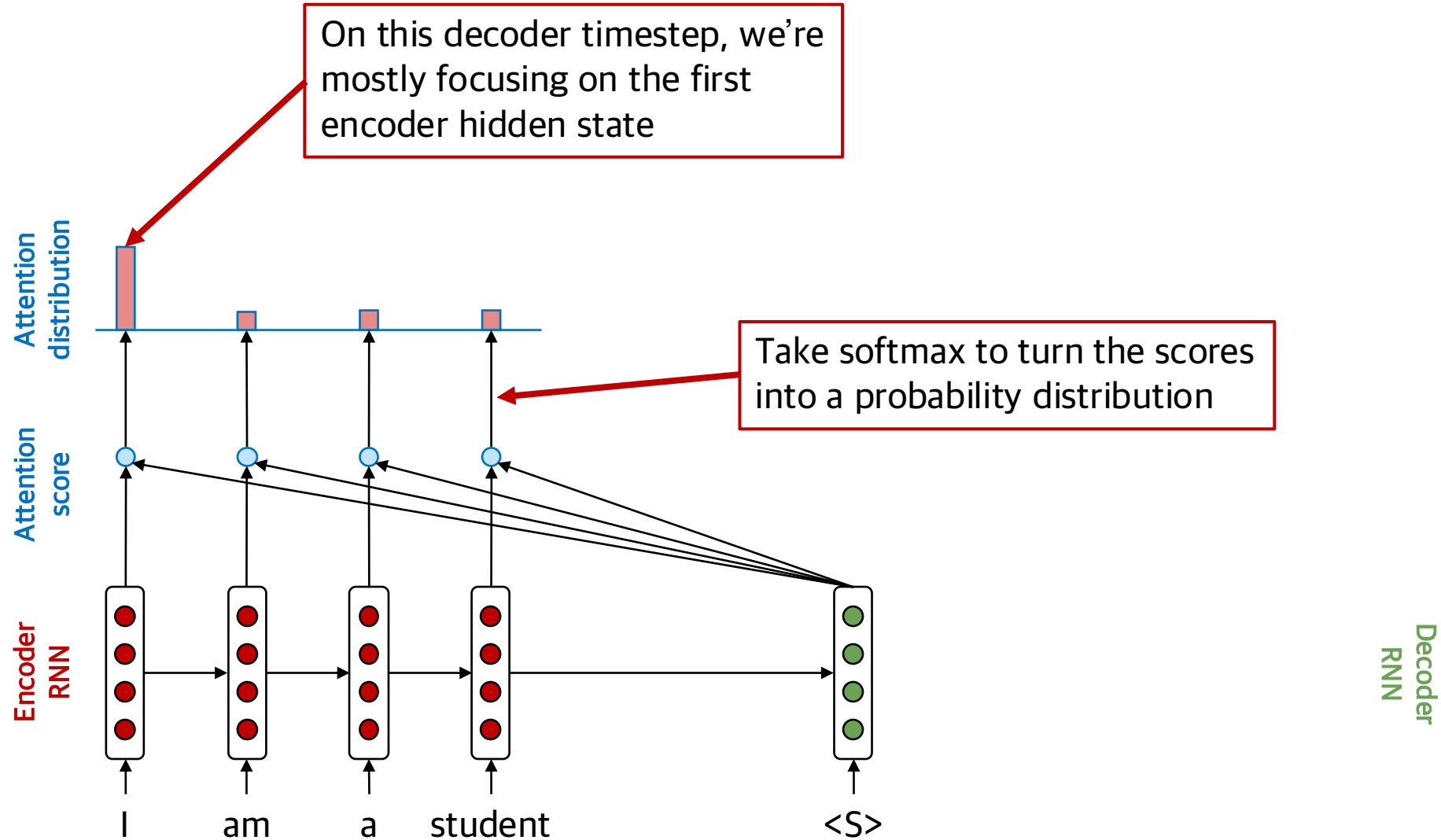
- Bottleneck problem에 대한 해결을 위해 **attention** 개념이 제안됨
- Core idea
 - 각 시점마다 decoder에서 입력 sequence의 특정 부분에 초점을 맞출 수 있도록 encoder로 직접적으로 연결

image from : <https://docs.likejazz.com/attention/>

5.2.2 Seq2Seq with Attention



5.2.2 Seq2Seq with Attention



5.2.2 참고 : Dot-Product

■ Inner Product

- vector $v = (x_1, x_2, \dots, x_n)$ 와 $w = (y_1, y_2, \dots, y_n)$ 의 inner product는

$$v \cdot w = v^T w = \sum_{i=1}^n x_i y_i$$

- Example

$$v = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, w = \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \text{일 때,}$$

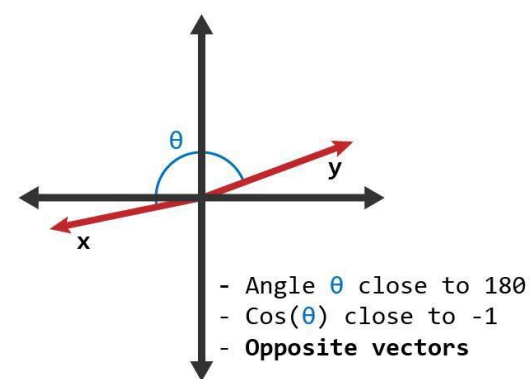
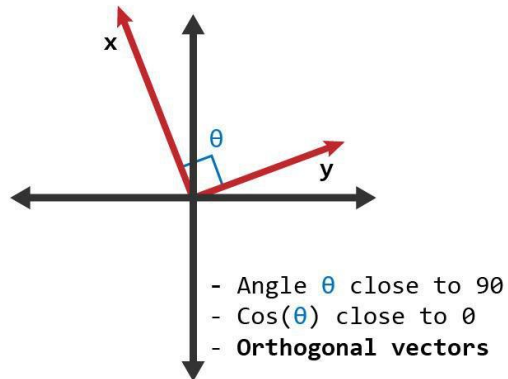
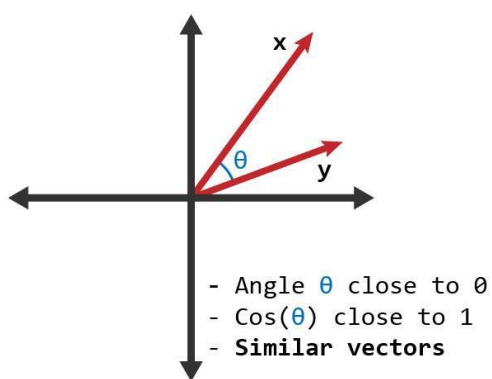
$$\begin{aligned} v \cdot w &= [1, 2, 0] * \begin{bmatrix} 2 \\ 0 \\ 2 \end{bmatrix} \\ &= 1 * 2 + (-1) * 0 + 0 * 2 = 2 + 0 + 0 = 2 \end{aligned}$$

5.2.2 참고 : Dot-Product

■ Cosine Similarity

- 두 vector v, w 에 대하여 그 사잇각을 θ ($0 \leq \theta \leq \pi$) 라고 하면

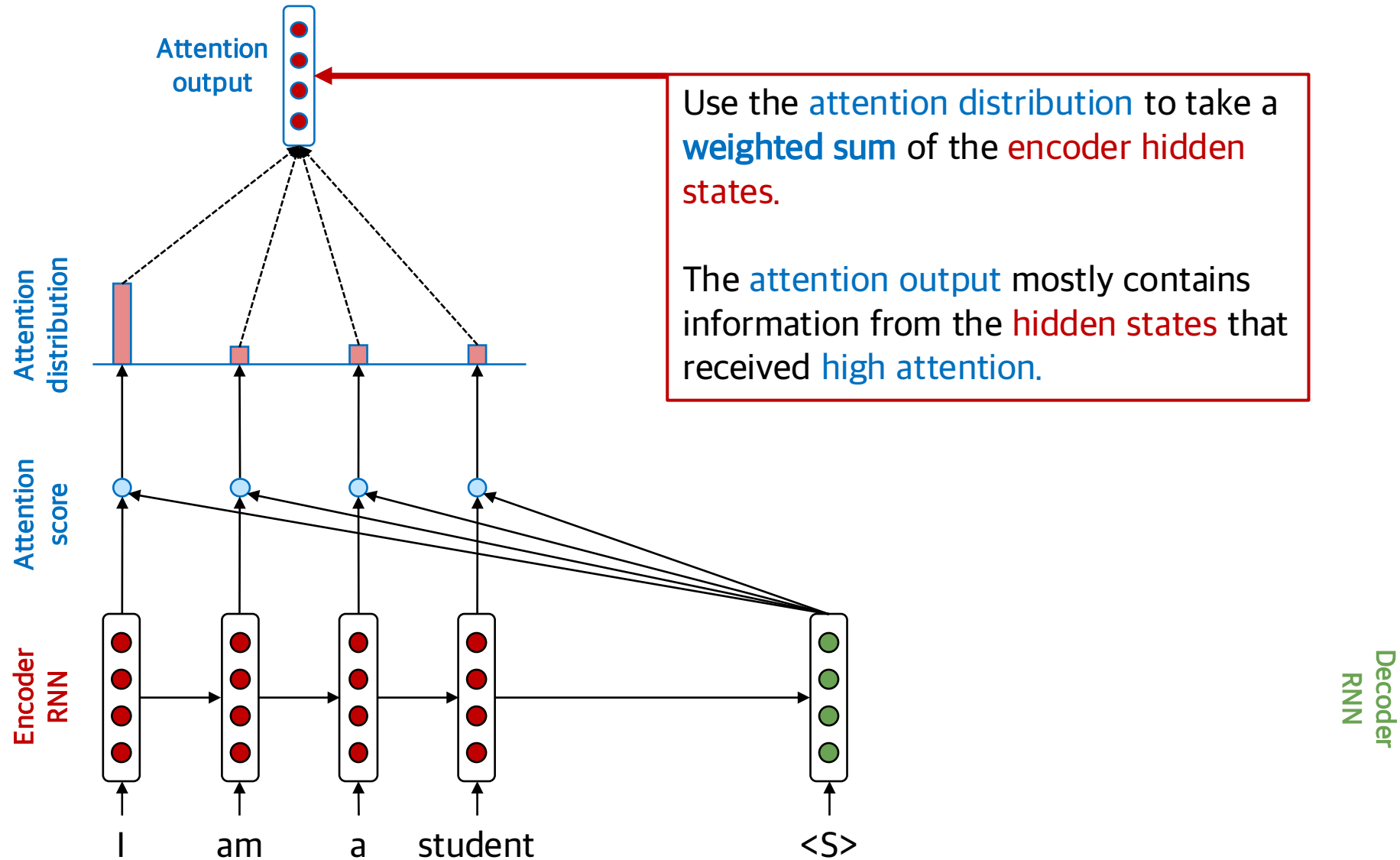
$$v \cdot w = \|v\| \|w\| \cos \theta, \quad \cos \theta = \frac{v \cdot w}{\|v\| \|w\|}$$



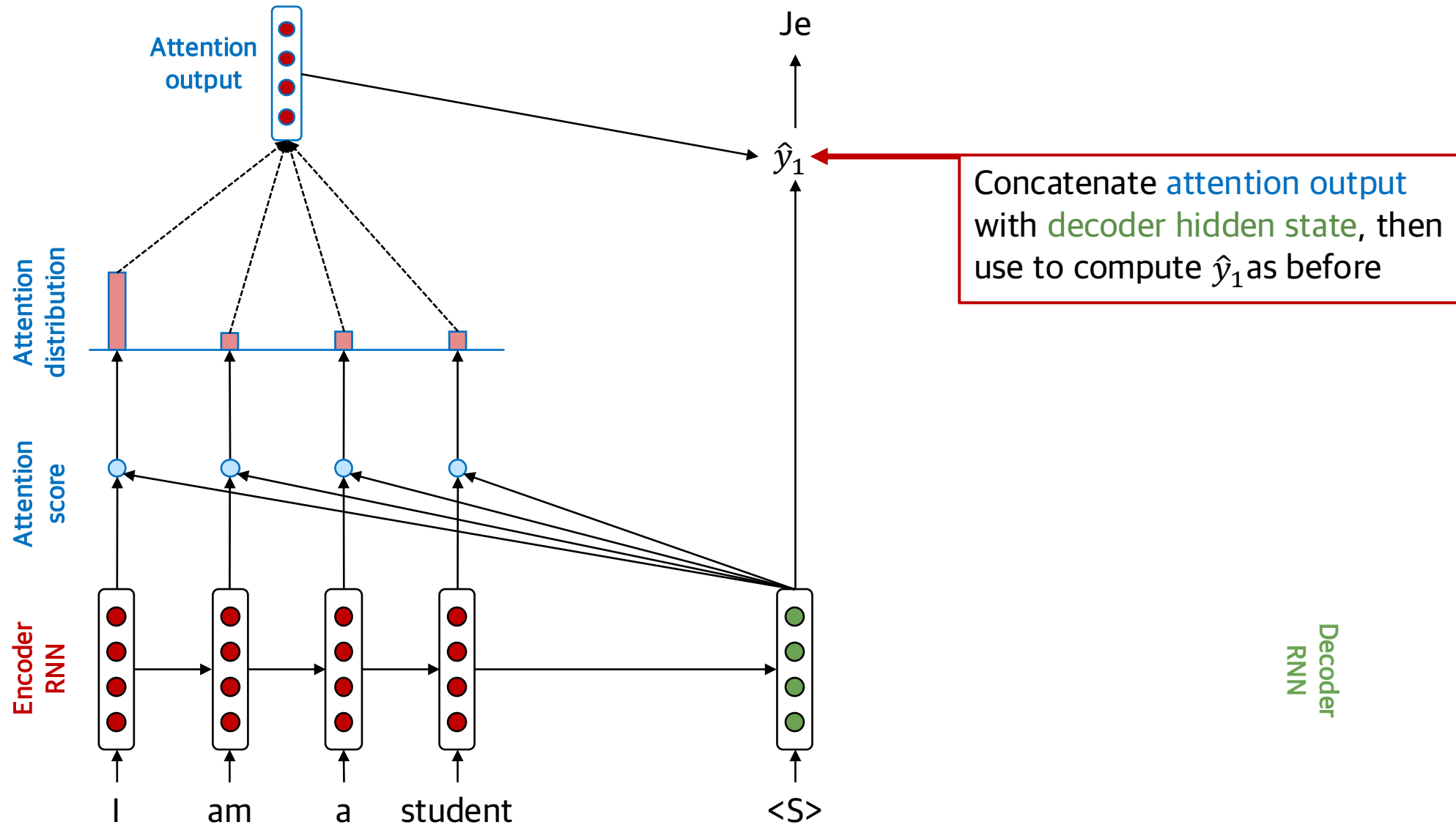
<https://www.leamdatasci.com/glossary/cosine-similarity/>

- $\cos \theta$ 값을 이용해 vector 간의 similarity 측정 가능
- 두 벡터의 방향이 동일할 경우 $\cos 0^\circ = 1$ 즉, $\cos \theta$ 가 1에 가까울수록 similarity가 높음

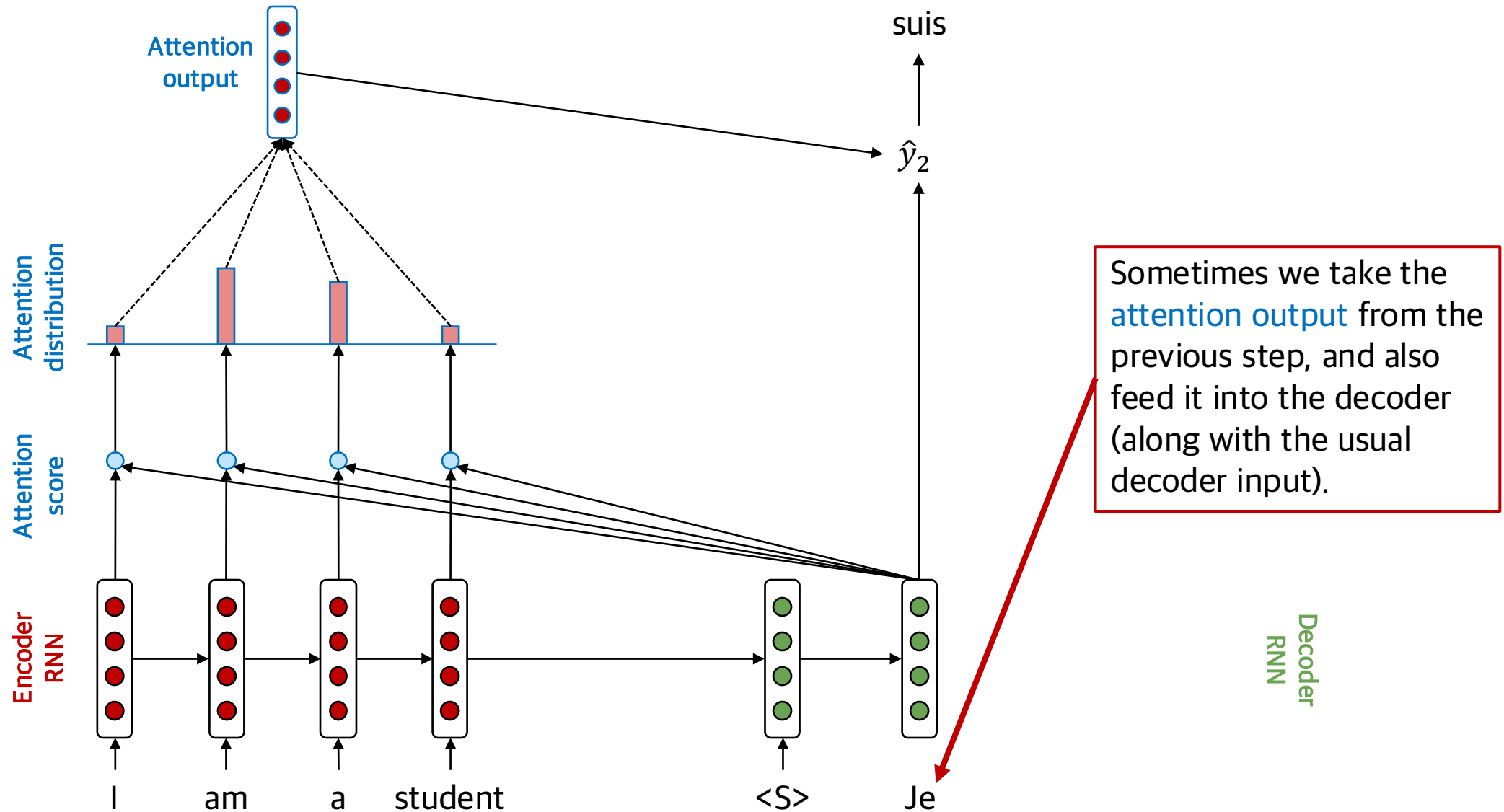
5.2.2 Seq2Seq with Attention



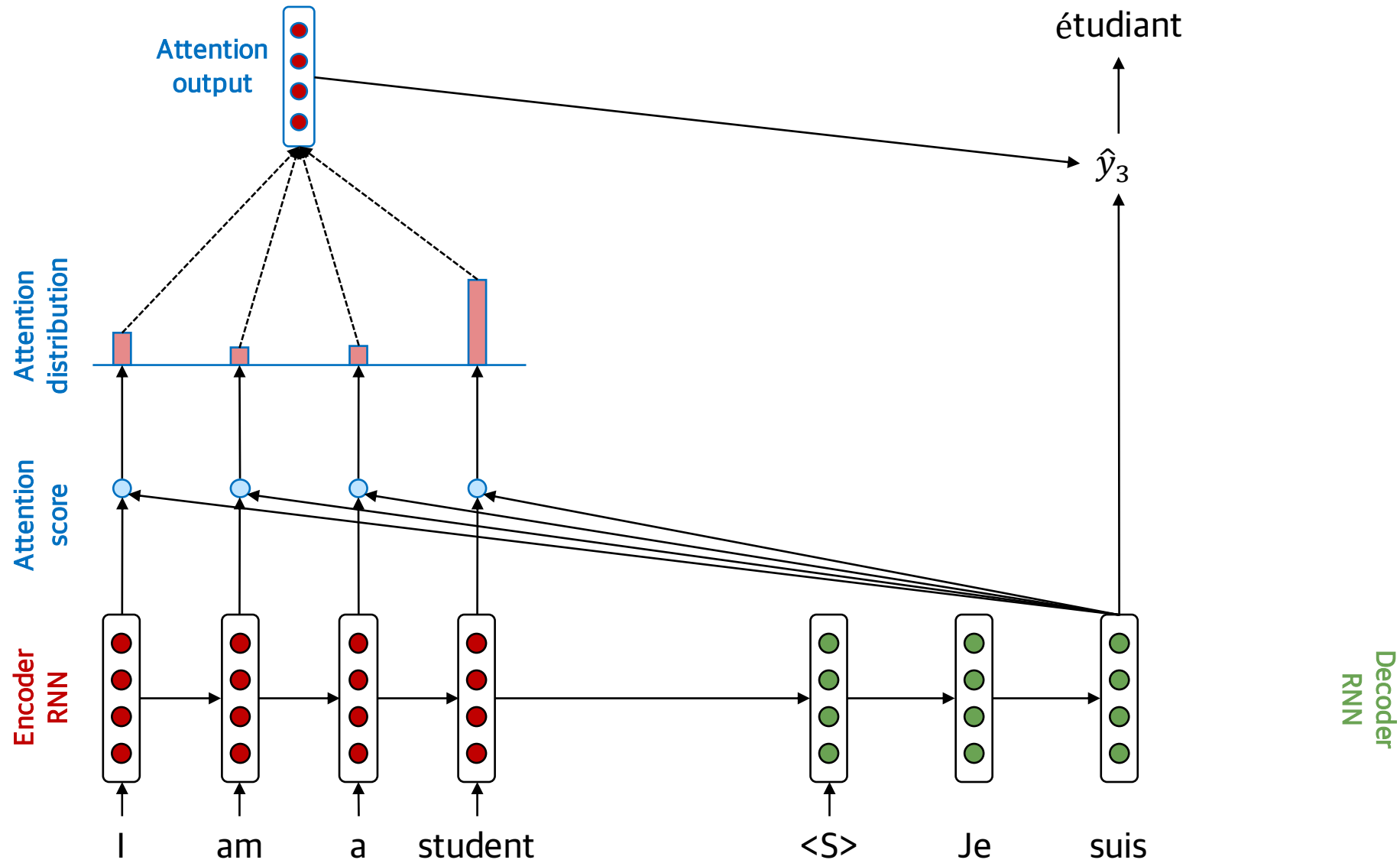
5.2.2 Seq2Seq with Attention



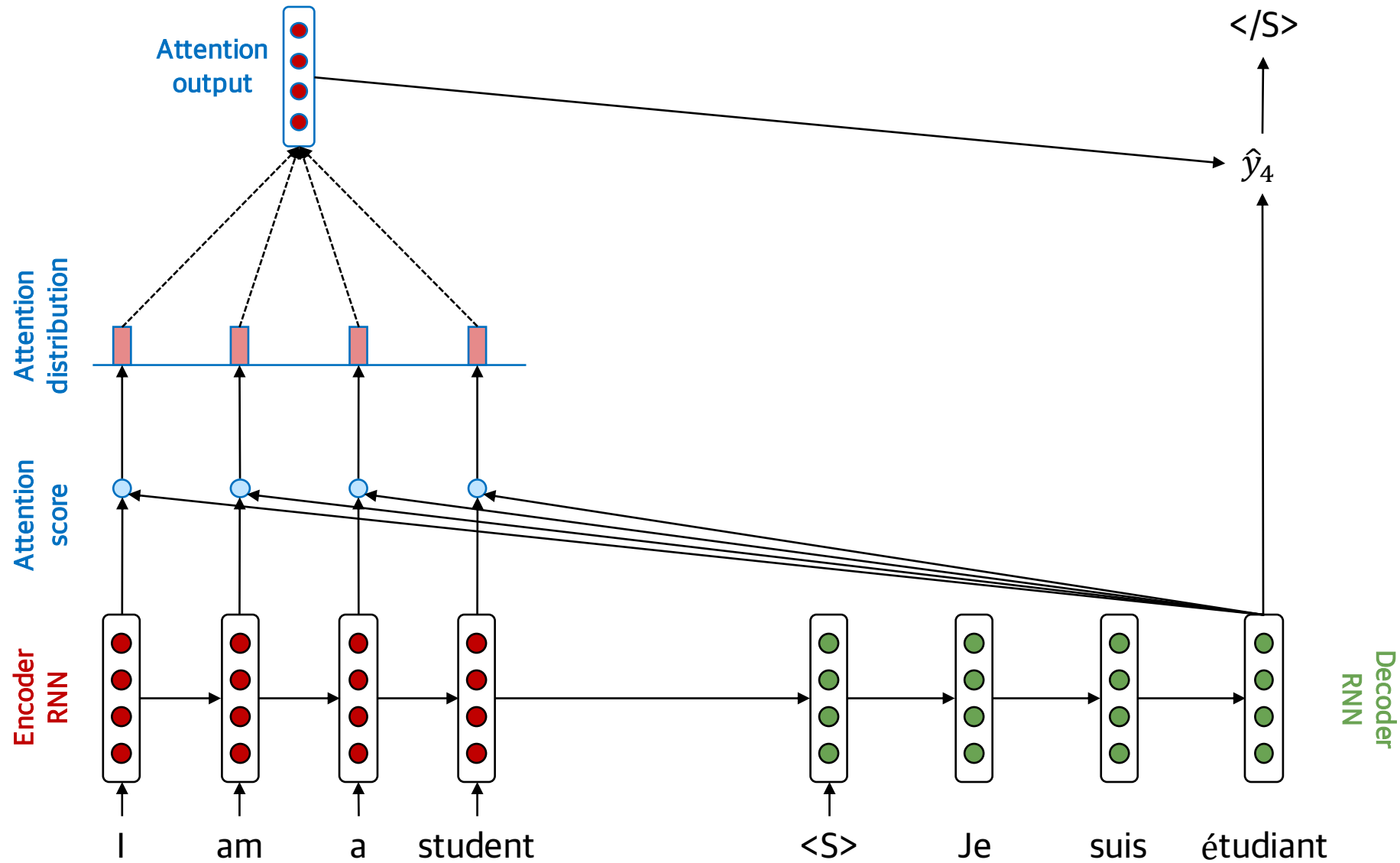
5.2.2 Seq2Seq with Attention



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5.2.2 Seq2Seq with Attention



5.2.2.1 Attention : in equations

- Encoder Hidden States $h_1, \dots, h_N \in \mathbb{R}^h$
- t 시점에서의 decoder hidden state $s_t \in \mathbb{R}^h$
- Attention score e^t 는 다음과 같이 계산

$$e^t = [s_t^T h_1, \dots, s_t^T h_N] \in \mathbb{R}^N$$

- Attention score에 softmax를 적용한 뒤, attention distribution α^t 를 계산
 - α^t 는 더해서 1이 되는 확률 분포

$$\alpha^t = \text{softmax}(e^t) \in \mathbb{R}^N$$

- Attention output a_t 은 α^t 와 encoder hidden state의 weighted sum을 통해서 계산

$$a_t = \sum_{i=1}^N \alpha_i^t h_i \in \mathbb{R}^h$$

- Attention output a_t 와 decoder hidden state s_t 를 concatenate한 뒤, 일반적인 seq2seq모델 처럼 처리

$$[a_t; s_t] \in \mathbb{R}^{2h}$$